Supplementary material for

Patient-level differential privacy, membership leakage and explanation fidelity in federated brain tumour MRI classification

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Overview

The supplement reports every configuration behind the main text, together with the controls and selection steps that the article summarises in a sentence. All tables were generated by `scripts/make_supplement.py` from the files frozen at repository tag `v0.9.1-results-frozen`, so each value can be traced to a hashed table or to a run of a frozen grid. Seeds 1 to 5 give every reported mean, seed 0 serves only for selection, and intervals are two-sided 95% t-intervals over seeds unless a caption states otherwise.

The sections follow the order of the article. Section S1 documents the duplicate check that defines the clean test set, Section S2 compares the full and the clean test sets and adds the source controls, Section S3 gives the privacy accounting and the selection costs, Section S4 covers client count and drift correction, Section S5 adds the membership-inference detail and Section S6 the explanation detail. Tables S5, S9 and S11 use the full test set of 813 images, and the clean-set values of the corresponding results come from Table S3.

S1 Benchmark curation and near-duplicate check

Perceptual hashing removed exact and near-exact copies during curation, yet it compares images in one orientation only. A second check therefore computed the normalised cross-correlation (NCC) between every pair of curated images at 64×64 pixels, once as stored and once with one image mirrored, keeping the larger value. Part A (Table S1) counts pairs above three thresholds and breaks the cross-split pairs down by the splits they join.

Table S1: Part A of the NCC check. Pairs of curated images (64×64) at or above each normalised cross-correlation threshold, taking the larger of the plain and the mirrored comparison.

NCC $\geq$	Pairs	Same unit	Cross-split	Test–train	Test–val	Train–val	Mirror only	Label conflict
0.9	2296	1665	295	190	29	76	139	19
0.95	682	357	148	84	17	47	120	0
0.98	195	28	80	43	9	28	79	0

Across the 813 test images, the NCC to the nearest image in another split has median 0.809, 90th percentile 0.926 and maximum 1.0000. 66 test images reach 0.95 and form the contaminated set.

Table S2: Part B of the NCC check. Agreement between perceptual-hash duplicates removed during curation and their kept partner, by Hamming distance d of the 64-bit hash.

d	Pairs	Median NCC	Min NCC	< 0.90	< 0.95	< 0.98
0	2945	1.0000	0.657	6	10	17
2	733	0.9992	0.629	13	22	82
4	296	0.9735	0.558	80	93	176

At the threshold of 0.95, fixed before the check, 148 pairs crossed a split boundary and 120 of them matched only after mirroring. Part B (Table S2) compares each duplicate removed during curation with the image it was merged into. Pairs removed at Hamming distance 0 and 2 are confirmed, with 19 of 3678 below NCC 0.90, whereas 80 of the 296 pairs at distance 4 fall below 0.90, many of them SARTAJ copies of Figshare slices with a different crop that NCC on whole images cannot align. Figure S1 shows the most similar cross-split pairs, most of them Br35H no-tumour images whose mirrored twin sat in another split.

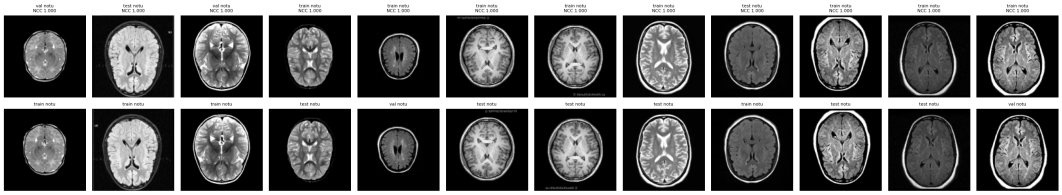


Figure S1: Cross-split pairs with the highest NCC after mirroring, each a test image and its nearest image in another split. Such pairs define the 66 contaminated test images.

S2 Full and clean test sets, and source controls

Table S3 lists balanced accuracy for every reported configuration on the full and on the clean test set. Removing the contaminated images changed configuration means by -0.049 to $+0.007$, mostly through lower no-tumour sensitivity once the mirrored Br35H twins left the test set, and no qualitative conclusion of the article changed.

Table S3: Balanced accuracy on the full test set (813 images) and on the clean test set (747 images) for every reported configuration, mean over seeds 1 to 5 with 95% t interval. sens_N is the no-tumour sensitivity.

Training	Algorithm	Regime	lr	Bal. acc. (full)	Bal. acc. (clean)	sens_N full	sens_N clean
central	–	no DP	0.02	0.919 (0.907, 0.931)	0.908 (0.894, 0.921)	0.958	0.913
central	–	clip only	0.02	0.871 (0.850, 0.891)	0.845 (0.818, 0.871)	0.902	0.800
central	–	$\varepsilon = 8$	0.02	0.815 (0.779, 0.851)	0.778 (0.739, 0.818)	0.846	0.700
central	–	$\varepsilon = 3$	0.02	0.686 (0.616, 0.756)	0.648 (0.572, 0.723)	0.738	0.587
central	–	$\varepsilon = 1$	0.02	0.331 (0.269, 0.393)	0.326 (0.272, 0.379)	0.138	0.117
central	–	patient clip only	0.02	0.884 (0.866, 0.903)	0.872 (0.852, 0.891)	0.954	0.903
central	–	patient $\varepsilon = 64$	0.02	0.836 (0.800, 0.873)	0.820 (0.778, 0.862)	0.938	0.873
central	–	patient $\varepsilon = 32$	0.02	0.745 (0.698, 0.792)	0.727 (0.682, 0.771)	0.922	0.850
central	–	patient $\varepsilon = 16$	0.02	0.584 (0.558, 0.611)	0.565 (0.543, 0.588)	0.902	0.827
central	–	patient $\varepsilon = 8$	0.02	0.353 (0.293, 0.413)	0.337 (0.284, 0.390)	0.890	0.823
central	–	patient $\varepsilon = 3$	0.02	0.253 (0.245, 0.261)	0.251 (0.248, 0.253)	0.992	0.983
Dir. 0.1, K=10	FedAvg	no DP	0.01	0.843 (0.834, 0.853)	0.817 (0.806, 0.828)	0.882	0.777
Dir. 0.1, K=10	FedAvg	$\varepsilon = 8$	0.003	0.677 (0.632, 0.722)	0.635 (0.588, 0.682)	0.614	0.447
Dir. 0.1, K=10	FedAvg	$\varepsilon = 3$	0.003	0.632 (0.586, 0.678)	0.589 (0.555, 0.624)	0.498	0.327
Dir. 0.1, K=10	FedAvg	$\varepsilon = 1$	0.003	0.322 (0.303, 0.341)	0.317 (0.301, 0.333)	0.197	0.177
Dir. 0.1, K=10	SCAFFOLD	no DP	0.01	0.815 (0.794, 0.836)	0.790 (0.773, 0.806)	0.901	0.800
Dir. 0.1, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.705 (0.682, 0.728)	0.661 (0.632, 0.690)	0.757	0.580
Dir. 0.1, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.633 (0.571, 0.696)	0.592 (0.530, 0.653)	0.549	0.383
Dir. 0.5, K=10	FedAvg	no DP	0.01	0.878 (0.863, 0.893)	0.870 (0.852, 0.887)	0.970	0.937
Dir. 0.5, K=10	FedAvg	$\varepsilon = 8$	0.003	0.718 (0.686, 0.750)	0.669 (0.626, 0.712)	0.771	0.577
Dir. 0.5, K=10	FedAvg	$\varepsilon = 8$	0.01	0.662 (0.586, 0.739)	0.624 (0.565, 0.684)	0.528	0.377
Dir. 0.5, K=10	FedAvg	$\varepsilon = 3$	0.003	0.626 (0.570, 0.682)	0.590 (0.548, 0.631)	0.550	0.403
Dir. 0.5, K=10	FedAvg	$\varepsilon = 1$	0.003	0.341 (0.286, 0.396)	0.331 (0.280, 0.383)	0.326	0.287
Dir. 0.5, K=10	SCAFFOLD	no DP	0.01	0.889 (0.871, 0.906)	0.877 (0.857, 0.897)	0.957	0.910
Dir. 0.5, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.713 (0.681, 0.744)	0.666 (0.623, 0.708)	0.784	0.597
Dir. 0.5, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.621 (0.560, 0.682)	0.582 (0.537, 0.627)	0.558	0.403
Dir. 0.5, K=15	FedAvg	no DP	0.01	0.903 (0.897, 0.909)	0.886 (0.874, 0.899)	0.938	0.870
Dir. 0.5, K=15	FedAvg	$\varepsilon = 8$	0.003	0.696 (0.618, 0.774)	0.655 (0.576, 0.735)	0.723	0.560
Dir. 0.5, K=15	FedAvg	$\varepsilon = 8$	0.01	0.564 (0.464, 0.664)	0.525 (0.449, 0.601)	0.435	0.280
Dir. 0.5, K=15	FedAvg	$\varepsilon = 3$	0.003	0.558 (0.458, 0.658)	0.525 (0.448, 0.602)	0.398	0.267
Dir. 0.5, K=3	FedAvg	no DP	0.01	0.868 (0.850, 0.886)	0.853 (0.827, 0.878)	0.917	0.857
Dir. 0.5, K=3	FedAvg	$\varepsilon = 8$	0.003	0.717 (0.644, 0.790)	0.675 (0.597, 0.754)	0.693	0.527
Dir. 0.5, K=3	FedAvg	$\varepsilon = 8$	0.01	0.820 (0.789, 0.852)	0.779 (0.743, 0.816)	0.794	0.630
Dir. 0.5, K=3	FedAvg	$\varepsilon = 3$	0.003	0.719 (0.698, 0.739)	0.677 (0.644, 0.710)	0.717	0.550
Dir. 0.5, K=5	FedAvg	no DP	0.01	0.906 (0.894, 0.919)	0.892 (0.879, 0.904)	0.946	0.887

Training	Algorithm	Regime	lr	Bal. acc. (full)	Bal. acc. (clean)	sens_N full	sens_N clean
Dir. 0.5, K=5	FedAvg	$\varepsilon = 8$	0.003	0.735 (0.712, 0.758)	0.694 (0.667, 0.721)	0.790	0.627
Dir. 0.5, K=5	FedAvg	$\varepsilon = 8$	0.01	0.773 (0.713, 0.833)	0.731 (0.665, 0.798)	0.779	0.613
Dir. 0.5, K=5	FedAvg	$\varepsilon = 3$	0.003	0.708 (0.684, 0.732)	0.669 (0.644, 0.693)	0.744	0.587
Dir. 5.0, K=10	FedAvg	no DP	0.01	0.899 (0.878, 0.919)	0.884 (0.858, 0.909)	0.946	0.887
Dir. 5.0, K=10	FedAvg	$\varepsilon = 8$	0.003	0.720 (0.695, 0.745)	0.674 (0.636, 0.712)	0.792	0.607
Dir. 5.0, K=10	FedAvg	$\varepsilon = 3$	0.003	0.623 (0.571, 0.676)	0.583 (0.542, 0.625)	0.550	0.390
Dir. 5.0, K=10	FedAvg	$\varepsilon = 1$	0.003	0.333 (0.280, 0.385)	0.328 (0.277, 0.379)	0.266	0.247
Dir. 5.0, K=10	SCAFFOLD	no DP	0.01	0.907 (0.897, 0.917)	0.892 (0.879, 0.905)	0.946	0.887
Dir. 5.0, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.716 (0.692, 0.740)	0.669 (0.632, 0.705)	0.789	0.600
Dir. 5.0, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.625 (0.563, 0.686)	0.585 (0.535, 0.635)	0.562	0.403
IID, K=10	FedAvg	no DP	0.01	0.904 (0.893, 0.914)	0.891 (0.879, 0.904)	0.954	0.903
IID, K=10	FedAvg	$\varepsilon = 8$	0.003	0.720 (0.698, 0.742)	0.673 (0.641, 0.704)	0.790	0.600
IID, K=10	FedAvg	$\varepsilon = 3$	0.003	0.639 (0.608, 0.671)	0.593 (0.558, 0.627)	0.606	0.420
IID, K=10	FedAvg	$\varepsilon = 1$	0.003	0.348 (0.311, 0.386)	0.344 (0.308, 0.380)	0.304	0.287
IID, K=10	SCAFFOLD	no DP	0.01	0.909 (0.892, 0.926)	0.896 (0.878, 0.914)	0.952	0.900
IID, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.719 (0.699, 0.740)	0.671 (0.642, 0.701)	0.789	0.597
IID, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.637 (0.605, 0.668)	0.589 (0.555, 0.623)	0.600	0.410
natural	FedAvg	no DP	0.01	0.875 (0.868, 0.882)	0.841 (0.833, 0.848)	0.824	0.687
natural	FedAvg	$\varepsilon = 8$	0.01	0.808 (0.783, 0.833)	0.765 (0.731, 0.800)	0.757	0.587
natural	FedAvg	$\varepsilon = 3$	0.003	0.714 (0.707, 0.721)	0.674 (0.659, 0.688)	0.750	0.590
natural	FedAvg	$\varepsilon = 1$	0.003	0.537 (0.474, 0.601)	0.525 (0.463, 0.586)	0.366	0.317
natural	FedAvg	weight noise $s = 0.5$	0.01	0.259 (0.235, 0.284)	0.254 (0.243, 0.266)	0.173	0.153
natural	FedAvg	weight noise $s = 0.003$	0.01	0.843 (0.826, 0.860)	0.801 (0.785, 0.816)	0.699	0.530
natural	FedAvg	patient $\varepsilon = 8$	0.003	0.403 (0.333, 0.472)	0.398 (0.330, 0.466)	0.123	0.103
natural	SCAFFOLD	no DP	0.01	0.900 (0.889, 0.911)	0.879 (0.863, 0.895)	0.920	0.837
natural	SCAFFOLD	$\varepsilon = 8$	0.01	0.825 (0.801, 0.849)	0.790 (0.757, 0.822)	0.837	0.697
natural	SCAFFOLD	$\varepsilon = 3$	0.003	0.713 (0.706, 0.719)	0.671 (0.661, 0.681)	0.770	0.603
unit Dir. 0.5, K=10	FedAvg	patient $\varepsilon = 8$	0.01	0.264 (0.234, 0.294)	0.271 (0.235, 0.307)	0.512	0.540

Local-only training (Table S4) shows what each client of the natural partition achieves alone. The Br35H client holds no tumour images and cannot exceed the chance level of 0.25, and even the large Figshare client reaches at most 0.625, below every non-private federated model.

Table S4: Local-only baselines. Each client of the natural partition trains alone on its own images with the central protocol and is tested on the shared test set (seed means).

Client	Regime	Seeds	Bal. acc. (full)	Bal. acc. (clean)
Br35H	$\varepsilon = 8$	3	0.249	0.255
Br35H	no DP	3	0.250	0.250
Figshare	$\varepsilon = 8$	3	0.464	0.464
Figshare	no DP	3	0.625	0.625
SARTAJ	$\varepsilon = 8$	3	0.245	0.250
SARTAJ	no DP	3	0.543	0.526

The per-source controls in Table S5 separate source composition from diagnostic skill. SARTAJ is the only source with all four classes, and its four-class balanced accuracy stays below the pooled value in every setting that learns more than chance. The pooled number therefore benefits from sources that hold only one or two classes.

Table S5: Per-source controls, full test set, seed means. SARTAJ has all four classes and Figshare only the three tumour classes. Specificity is no-tumour recall within a source.

Mode	Partition	Regime	Bal. acc.	SARTAJ (4 cl.)	Figshare (3 cl.)	Tumour sens.	Spec. SARTAJ	Spec. Br35H
central	natural	no DP	0.919	0.663	0.924	0.984	0.919	0.994
central	natural	clip only	0.871	0.586	0.885	0.975	0.797	0.997
central	natural	$\varepsilon = 8$	0.815	0.528	0.830	0.975	0.702	0.976
central	natural	$\varepsilon = 3$	0.686	0.467	0.681	0.958	0.590	0.870
central	natural	$\varepsilon = 1$	0.331	0.326	0.394	0.967	0.125	0.148
central	natural	patient clip only	0.884	0.636	0.882	0.974	0.902	1.000
central	natural	patient $\varepsilon = 64$	0.836	0.617	0.818	0.962	0.871	0.997
central	natural	patient $\varepsilon = 32$	0.745	0.561	0.695	0.960	0.847	0.988
central	natural	patient $\varepsilon = 16$	0.584	0.449	0.487	0.845	0.820	0.976
central	natural	patient $\varepsilon = 8$	0.353	0.290	0.177	0.462	0.824	0.948
central	natural	patient $\varepsilon = 3$	0.253	0.256	0.006	0.018	0.983	1.000

Mode	Partition	Regime	Bal. acc.	SARTAJ (4 cl.)	Figshare (3 cl.)	Tumour sens.	Spec. SARTAJ	Spec. Br35H
federated	Dir. 0.1, K=10	no DP	0.843	0.594	0.845	0.985	0.769	0.982
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.677	0.441	0.712	0.979	0.434	0.776
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.632	0.429	0.690	0.972	0.342	0.636
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.322	0.317	0.359	0.948	0.183	0.209
federated	Dir. 0.5, K=10	no DP	0.878	0.676	0.861	0.984	0.936	1.000
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.718	0.465	0.716	0.970	0.576	0.945
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.626	0.434	0.662	0.973	0.397	0.688
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.341	0.326	0.345	0.878	0.292	0.358
federated	Dir. 5.0, K=10	no DP	0.899	0.631	0.903	0.986	0.885	1.000
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.720	0.462	0.714	0.968	0.607	0.958
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.623	0.428	0.658	0.975	0.393	0.691
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.333	0.326	0.354	0.901	0.247	0.282
federated	IID, K=10	no DP	0.904	0.650	0.906	0.985	0.902	1.000
federated	IID, K=10	$\varepsilon = 8$	0.720	0.464	0.714	0.967	0.600	0.961
federated	IID, K=10	$\varepsilon = 3$	0.639	0.432	0.661	0.967	0.427	0.767
federated	IID, K=10	$\varepsilon = 1$	0.348	0.335	0.363	0.879	0.288	0.318
federated	natural	no DP	0.875	0.599	0.913	0.990	0.678	0.955
federated	natural	$\varepsilon = 8$	0.808	0.517	0.852	0.976	0.590	0.906
federated	natural	$\varepsilon = 3$	0.714	0.498	0.716	0.977	0.593	0.891
federated	natural	$\varepsilon = 1$	0.537	0.357	0.603	0.965	0.312	0.415
federated	natural	patient $\varepsilon = 8$	0.403	0.312	0.503	0.967	0.092	0.152
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.264	0.255	0.184	0.497	0.549	0.479

The probe in Table S6 tests whether penultimate features reveal the source collection within one class. Trained models decode the source only slightly better than an ImageNet-initialised network and a random network, so the source signal is already present in the images before training.

Table S6: Source probe. Balanced accuracy of a logistic-regression probe on penultimate features that predicts the source within one class (tumour, Figshare against SARTAJ. No tumour, SARTAJ against Br35H).

Mode	Partition	Regime	Tumour probe	No-tumour probe
central	natural	no DP	0.913 (0.888, 0.938)	0.821 (0.812, 0.831)
central	natural	clip only	0.908 (0.878, 0.938)	0.808 (0.795, 0.821)
central	natural	$\varepsilon = 8$	0.896 (0.882, 0.909)	0.794 (0.757, 0.830)
central	natural	$\varepsilon = 3$	0.882 (0.824, 0.941)	0.770 (0.735, 0.806)
central	natural	$\varepsilon = 1$	0.851 (0.821, 0.882)	0.769 (0.699, 0.839)
central	natural	patient clip only	0.914 (0.889, 0.939)	0.813 (0.791, 0.835)
central	natural	patient $\varepsilon = 64$	0.876 (0.857, 0.895)	0.812 (0.781, 0.843)
central	natural	patient $\varepsilon = 32$	0.864 (0.832, 0.895)	0.762 (0.730, 0.793)
central	natural	patient $\varepsilon = 16$	0.827 (0.798, 0.855)	0.780 (0.744, 0.816)
central	natural	patient $\varepsilon = 8$	0.828 (0.784, 0.871)	0.772 (0.747, 0.797)
central	natural	patient $\varepsilon = 3$	0.794 (0.736, 0.851)	0.753 (0.699, 0.807)
federated	Dir. 0.1, K=10	no DP	0.920 (0.886, 0.955)	0.827 (0.814, 0.841)
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.895 (0.863, 0.927)	0.807 (0.790, 0.825)
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.891 (0.862, 0.920)	0.781 (0.758, 0.805)
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.868 (0.827, 0.909)	0.812 (0.763, 0.861)
federated	Dir. 0.5, K=10	no DP	0.930 (0.919, 0.940)	0.831 (0.814, 0.849)
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.901 (0.870, 0.932)	0.813 (0.789, 0.836)
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.891 (0.861, 0.922)	0.814 (0.765, 0.863)
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.859 (0.821, 0.898)	0.791 (0.756, 0.826)
federated	Dir. 5.0, K=10	no DP	0.921 (0.895, 0.946)	0.825 (0.804, 0.846)
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.893 (0.875, 0.911)	0.800 (0.765, 0.834)
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.893 (0.859, 0.926)	0.813 (0.758, 0.868)
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.872 (0.842, 0.901)	0.797 (0.764, 0.830)
federated	IID, K=10	no DP	0.931 (0.906, 0.955)	0.839 (0.821, 0.857)
federated	IID, K=10	$\varepsilon = 8$	0.897 (0.872, 0.921)	0.813 (0.781, 0.844)
federated	IID, K=10	$\varepsilon = 3$	0.883 (0.859, 0.907)	0.821 (0.769, 0.873)
federated	IID, K=10	$\varepsilon = 1$	0.865 (0.822, 0.909)	0.775 (0.745, 0.805)
federated	natural	no DP	0.903 (0.890, 0.916)	0.808 (0.755, 0.861)
federated	natural	$\varepsilon = 8$	0.871 (0.845, 0.898)	0.811 (0.782, 0.841)
federated	natural	$\varepsilon = 3$	0.875 (0.845, 0.904)	0.799 (0.776, 0.823)
federated	natural	$\varepsilon = 1$	0.847 (0.809, 0.886)	0.790 (0.772, 0.807)
federated	natural	patient $\varepsilon = 8$	0.840 (0.782, 0.898)	0.799 (0.706, 0.892)
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.822 (0.769, 0.876)	0.759 (0.727, 0.790)

Mode	Partition	Regime	Tumour probe	No-tumour probe
baseline	–	ImageNet initialisation	0.871	0.783
baseline	–	random network	0.846	0.776

S3 Privacy accounting and selection cost

Table S7 gives the accounted privacy loss of every private configuration. Federated learning rates were chosen among two candidates trained on the same records, and composing the guarantee over both candidates raises record-level $\varepsilon = 8$ to 11.98, following the analysis of Papernot and Steinke for hyperparameter search.

Table S7: Accounted privacy loss ($\delta = 10^{-5}$, 1000 steps, $q = 0.1$). $\varepsilon_{k=2}$ composes over the two learning-rate candidates, and the worst-unit ε over the clients holding one unit.

Mode	Partition	Unit	Target ε	σ	ε	$\varepsilon_{k=2}$	Worst-unit ε
central	natural	image	1	12.885	1.00	–	–
central	natural	image	3	4.832	3.00	–	–
central	natural	patient	3	4.832	3.00	–	3.00
central	natural	image	8	2.179	7.96	–	–
central	natural	patient	8	2.179	7.96	–	7.96
central	natural	patient	16	1.353	15.96	–	15.96
central	natural	patient	32	0.922	31.90	–	31.90
central	natural	patient	64	0.673	63.69	–	63.69
federated	Dir. 0.1, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 0.1, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 0.1, K=10	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 0.5, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=10	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=15	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=15	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=3	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=3	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=5	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=5	image	8	2.179	7.96	11.98	–
federated	Dir. 5.0, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 5.0, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 5.0, K=10	image	8	2.179	7.96	11.98	–
federated	IID, K=10	image	1	12.885	1.00	1.46	–
federated	IID, K=10	image	3	4.832	3.00	4.42	–
federated	IID, K=10	image	8	2.179	7.96	11.98	–
federated	natural	image	1	12.885	1.00	1.46	–
federated	natural	image	3	4.832	3.00	4.42	–
federated	natural	image	8	2.179	7.96	11.98	–
federated	natural	patient	8	2.179	7.96	18.39	11.98
federated	unit Dir. 0.5, K=10	image	3	4.832	3.00	4.42	–
federated	unit Dir. 0.5, K=10	image	8	2.179	7.96	11.98	–
federated	unit Dir. 0.5, K=10	patient	8	2.179	7.96	11.98	7.96

Central models use the learning rate 0.02 fixed in pilot runs on the same training data, so their selection cost is disclosed but not composed. Natural3 splits 11 units across two clients, which raises the worst-unit guarantee of that partition.

Table S8 lists the learning-rate candidates at seed 0 and the one selected by lowest validation loss. Every non-private configuration selected 0.01, while most private ones selected 0.003, consistent with the smaller steps that noisy gradients tolerate.

Table S8: Learning-rate selection at seed 0 for federated models, by lowest best validation loss among two candidates. Seed 0 is never used for reported results.

Partition	Regime	lr	Best val. loss	Val. bal. acc.	Selected
natural	no DP	0.003	0.447	0.801	
natural	no DP	0.01	0.333	0.847	✓

Partition	Regime	lr	Best val. loss	Val. bal. acc.	Selected
IID, K=10	no DP	0.003	0.310	0.882	
IID, K=10	no DP	0.01	0.255	0.911	✓
Dir. 0.1, K=10	no DP	0.003	0.542	0.720	
Dir. 0.1, K=10	no DP	0.01	0.378	0.838	✓
Dir. 0.5, K=10	no DP	0.003	0.356	0.836	
Dir. 0.5, K=10	no DP	0.01	0.318	0.889	✓
Dir. 5.0, K=10	no DP	0.003	0.309	0.875	
Dir. 5.0, K=10	no DP	0.01	0.267	0.896	✓
natural	$\varepsilon = 8$	0.003	0.780	0.678	
natural	$\varepsilon = 8$	0.01	0.656	0.811	✓
IID, K=10	$\varepsilon = 8$	0.003	0.784	0.691	✓
IID, K=10	$\varepsilon = 8$	0.01	0.823	0.648	
Dir. 0.1, K=10	$\varepsilon = 8$	0.003	0.751	0.689	✓
Dir. 0.1, K=10	$\varepsilon = 8$	0.01	0.901	0.568	
Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.763	0.692	✓
Dir. 0.5, K=10	$\varepsilon = 8$	0.01	0.877	0.644	
Dir. 5.0, K=10	$\varepsilon = 8$	0.003	0.779	0.685	✓
Dir. 5.0, K=10	$\varepsilon = 8$	0.01	0.835	0.625	
natural	$\varepsilon = 3$	0.003	0.784	0.663	✓
natural	$\varepsilon = 3$	0.01	0.812	0.588	
IID, K=10	$\varepsilon = 3$	0.003	0.928	0.504	✓
IID, K=10	$\varepsilon = 3$	0.01	1.228	0.336	
Dir. 0.1, K=10	$\varepsilon = 3$	0.003	0.957	0.521	✓
Dir. 0.1, K=10	$\varepsilon = 3$	0.01	1.241	0.299	
Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.970	0.560	✓
Dir. 0.5, K=10	$\varepsilon = 3$	0.01	1.255	0.299	
Dir. 5.0, K=10	$\varepsilon = 3$	0.003	0.953	0.483	✓
Dir. 5.0, K=10	$\varepsilon = 3$	0.01	1.233	0.308	
natural	$\varepsilon = 1$	0.003	0.980	0.518	✓
natural	$\varepsilon = 1$	0.01	1.363	0.274	
IID, K=10	$\varepsilon = 1$	0.003	1.307	0.389	✓
IID, K=10	$\varepsilon = 1$	0.01	1.769	0.275	
Dir. 0.1, K=10	$\varepsilon = 1$	0.003	1.243	0.372	✓
Dir. 0.1, K=10	$\varepsilon = 1$	0.01	1.889	0.251	
Dir. 0.5, K=10	$\varepsilon = 1$	0.003	1.311	0.384	✓
Dir. 0.5, K=10	$\varepsilon = 1$	0.01	1.738	0.269	
Dir. 5.0, K=10	$\varepsilon = 1$	0.003	1.311	0.380	✓
Dir. 5.0, K=10	$\varepsilon = 1$	0.01	1.693	0.276	
natural	patient $\varepsilon = 8$	0.003	1.191	0.408	✓
natural	patient $\varepsilon = 8$	0.01	1.543	0.280	
unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.003	2.098	0.353	
unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.01	2.051	0.246	✓
natural	patient $\varepsilon = 3$	0.003	1.410	0.267	✓
natural	patient $\varepsilon = 3$	0.01	15.649	0.281	
unit Dir. 0.5, K=10	patient $\varepsilon = 3$	0.003	2.275	0.214	✓
unit Dir. 0.5, K=10	patient $\varepsilon = 3$	0.01	15.845	0.250	
unit Dir. 0.5, K=10	no DP	0.003	0.385	0.843	
unit Dir. 0.5, K=10	no DP	0.01	0.306	0.886	✓
unit Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.783	0.681	✓
unit Dir. 0.5, K=10	$\varepsilon = 8$	0.01	0.911	0.520	
unit Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.958	0.599	✓
unit Dir. 0.5, K=10	$\varepsilon = 3$	0.01	1.292	0.322	

S4 Client count and drift correction

Table S9 holds the client-count experiment. The aggregated noise standard deviation grows with $\sqrt{K}$ at a fixed noise multiplier, and accuracy at $\varepsilon = 3$ falls as it grows, whereas accuracy without privacy shows no trend.

Table S9: Client count at fixed data, FedAvg on Dirichlet ($\alpha = 0.5$) partitions, full test set, learning rate 0.003 under DP and 0.01 without. Aggregated noise s.d. is $\sigma C\sqrt{K}/(qN)$, $N = 2610$.

K	Regime	Client size min/med./max	σ	Agg. noise s.d.	Bal. acc.	sens$_M$
3	$\varepsilon = 3$	424/618/1568	4.832	0.0321	0.719 (0.698, 0.739)	0.315
5	$\varepsilon = 3$	204/452/1065	4.832	0.0414	0.708 (0.684, 0.732)	0.171
10	$\varepsilon = 3$	82/225/533	4.832	0.0585	0.626 (0.570, 0.682)	0.067
15	$\varepsilon = 3$	29/190/354	4.832	0.0717	0.558 (0.458, 0.658)	0.030
3	$\varepsilon = 8$	424/618/1568	2.179	0.0145	0.717 (0.644, 0.790)	0.305
5	$\varepsilon = 8$	204/452/1065	2.179	0.0187	0.735 (0.712, 0.758)	0.244
10	$\varepsilon = 8$	82/225/533	2.179	0.0264	0.718 (0.686, 0.750)	0.193
15	$\varepsilon = 8$	29/190/354	2.179	0.0323	0.696 (0.618, 0.774)	0.163
3	no DP	424/618/1568	0.000	0.0000	0.868 (0.850, 0.886)	0.643
5	no DP	204/452/1065	0.000	0.0000	0.906 (0.894, 0.919)	0.811
10	no DP	82/225/533	0.000	0.0000	0.878 (0.863, 0.893)	0.605
15	no DP	29/190/354	0.000	0.0000	0.903 (0.897, 0.909)	0.799

FedProx screening at seed 0 (Table S10) shows that proximal weights of 0.01 and 0.1 change validation loss by at most 0.0098. Larger weights lowered the loss in a few private settings and raised it elsewhere, without a consistent pattern that would justify carrying FedProx to the seed runs.

Table S10: FedProx screening at seed 0. Best validation loss of FedAvg ($\mu = 0$) and of FedProx with proximal weight μ, at the learning rate selected for FedAvg.

Partition	Regime	lr	$\mu = 0$	$\mu = 0.01$	$\mu = 0.1$	$\mu = 1$	$\mu = 10$
Dir. 0.1, K=10	$\varepsilon = 3$	0.003	0.957	0.957	0.954	0.944	0.930
Dir. 0.1, K=10	$\varepsilon = 8$	0.003	0.751	0.751	0.751	0.751	0.770
Dir. 0.1, K=10	no DP	0.01	0.378	0.374	0.378	0.405	0.570
Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.970	0.970	0.969	0.962	0.906
Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.763	0.763	0.764	0.767	0.812
Dir. 0.5, K=10	no DP	0.01	0.318	0.318	0.317	0.323	0.339
Dir. 5.0, K=10	$\varepsilon = 3$	0.003	0.953	0.953	0.952	0.942	0.911
Dir. 5.0, K=10	$\varepsilon = 8$	0.003	0.779	0.779	0.779	0.783	0.819
Dir. 5.0, K=10	no DP	0.01	0.267	0.267	0.266	0.266	0.300
IID, K=10	$\varepsilon = 3$	0.003	0.928	0.928	0.928	0.924	0.900
IID, K=10	$\varepsilon = 8$	0.003	0.784	0.784	0.784	0.786	0.823
IID, K=10	no DP	0.01	0.255	0.255	0.254	0.262	0.290
natural	$\varepsilon = 3$	0.003	0.784	0.784	0.785	0.789	0.838
natural	$\varepsilon = 8$	0.01	0.656	0.658	0.656	0.665	0.734
natural	no DP	0.01	0.333	0.340	0.343	0.371	0.516

Table S11 compares size-weighted SCAFFOLD with FedAvg. No difference survives Holm correction, and the uniform server average of the original algorithm collapses on unequal clients under privacy, most visibly on the natural partition, where one client holds 2029 of the 2610 training images (77.7%).

Table S11: SCAFFOLD against FedAvg, full test set. Size-weighted SCAFFOLD over seeds 1 to 5 with paired t-tests and Holm correction, and the uniform server average over seeds 1 to 3.

Partition	Regime	FedAvg	SCAFFOLD (size)	Diff.	p	p Holm	SCAFFOLD (uniform)
natural	no DP	0.875	0.900	+0.025	0.0043	0.064	0.898
Dir. 5.0, K=10	$\varepsilon = 8$	0.720	0.716	-0.004	0.0054	0.075	0.713
Dir. 0.1, K=10	no DP	0.843	0.815	-0.028	0.0132	0.172	0.761
natural	$\varepsilon = 8$	0.808	0.825	+0.017	0.0136	0.172	0.590
Dir. 0.5, K=10	no DP	0.878	0.889	+0.011	0.0514	0.566	0.886
Dir. 0.1, K=10	$\varepsilon = 8$	0.677	0.705	+0.028	0.1132	1.000	0.428
IID, K=10	no DP	0.904	0.909	+0.005	0.2159	1.000	0.904
IID, K=10	$\varepsilon = 3$	0.639	0.637	-0.003	0.2306	1.000	0.642
Dir. 0.5, K=10	$\varepsilon = 3$	0.626	0.621	-0.006	0.3204	1.000	0.492
Dir. 5.0, K=10	no DP	0.898	0.907	+0.008	0.3871	1.000	0.903
Dir. 0.5, K=10	$\varepsilon = 8$	0.718	0.713	-0.005	0.4677	1.000	0.718
IID, K=10	$\varepsilon = 8$	0.720	0.719	-0.001	0.5066	1.000	0.728
natural	$\varepsilon = 3$	0.714	0.713	-0.001	0.7524	1.000	0.358
Dir. 5.0, K=10	$\varepsilon = 3$	0.623	0.625	+0.001	0.8260	1.000	0.642
Dir. 0.1, K=10	$\varepsilon = 3$	0.632	0.633	+0.001	0.9456	1.000	0.268

S5 Membership inference

The weight-noise comparator used a rule fixed before any seed run. Table S12 shows that the published $s = 0.5$ diverged, and that the rule selected $s = 0.003$ as the largest level whose validation loss did not exceed that of record-level $\varepsilon = 8$.

Table S12: Selection of the utility-matched weight-noise level s at seed 0 on the natural partition. Noise $\mathcal{N}(0, s^2)$ is added to client weights without clipping.

s	Diverged	Best val. loss	Meets reference
0.5	yes	21.214	no
0.05	no	15.452	no
0.01	no	0.945	no
0.003	no	0.357	yes
0.001	no	0.313	yes

The reference is the seed-0 validation loss of record-level $\varepsilon = 8$ FedAvg on the natural partition, 0.656. The rule takes the largest s that meets it. Over seeds 1 to 5 on the full test set, weight noise $s = 0.5$ reaches balanced accuracy 0.259 (0.235, 0.284) and weight noise $s = 0.003$ reaches balanced accuracy 0.843 (0.826, 0.860).

Tables S13 and S14 report LiRA with the clean and with the full test set as non-members, in its online and its offline form. The choice of non-member set changes no conclusion, since online AUC, true-positive rate at 1% false positives and patient AUC move by less than 0.01 for every target that produced a usable model, and every private setting stays at the level of a random guess under both definitions.

Table S13: LiRA against natural3 FedAvg with 16 shadow models, clean test set (747 images) as non-members. Online and offline variants, mean over target seeds with 95% t interval.

Target	n	AUC	TPR@1%	TPR@0.1%	Patient AUC	Off. AUC	Off. TPR@1%	Off. patient AUC
no DP	5	0.666 (0.642, 0.689)	0.108 (0.081, 0.135)	0.033 (0.019, 0.046)	0.805 (0.762, 0.848)	0.656	0.068	0.769
$\varepsilon = 8$	5	0.464 (0.448, 0.480)	0.017 (0.012, 0.022)	0.006 (0.003, 0.008)	0.454 (0.427, 0.481)	0.503	0.016	0.511
$\varepsilon = 3$	5	0.545 (0.522, 0.567)	0.015 (0.008, 0.023)	0.003 (0.002, 0.005)	0.567 (0.529, 0.605)	0.546	0.014	0.581
$\varepsilon = 1$	5	0.479 (0.469, 0.489)	0.010 (0.006, 0.015)	0.003 (0.000, 0.005)	0.457 (0.413, 0.500)	0.488	0.012	0.500
weight noise $s = 0.5$	5	0.474 (0.437, 0.511)	0.038 (0.023, 0.053)	0.003 (0.002, 0.003)	0.518 (0.497, 0.539)	0.538	0.034	0.545
weight noise $s = 0.003$	5	0.594 (0.572, 0.616)	0.045 (0.038, 0.051)	0.016 (0.012, 0.020)	0.745 (0.713, 0.778)	0.569	0.030	0.665

Table S14: LiRA against natural3 FedAvg with 16 shadow models, full test set (813 images) as non-members. Online and offline variants, mean over target seeds with 95% t interval.

Target	n	AUC	TPR@1%	TPR@0.1%	Patient AUC	Off. AUC	Off. TPR@1%	Off. patient AUC
no DP	5	0.662 (0.642, 0.682)	0.112 (0.084, 0.140)	0.033 (0.019, 0.046)	0.805 (0.762, 0.848)	0.661	0.069	0.769
$\varepsilon = 8$	5	0.472 (0.459, 0.484)	0.015 (0.010, 0.020)	0.002 (-0.000, 0.005)	0.454 (0.427, 0.481)	0.497	0.013	0.511
$\varepsilon = 3$	5	0.545 (0.524, 0.565)	0.017 (0.010, 0.023)	0.003 (0.002, 0.005)	0.567 (0.529, 0.604)	0.546	0.014	0.581
$\varepsilon = 1$	5	0.483 (0.473, 0.493)	0.009 (0.005, 0.013)	0.001 (-0.000, 0.002)	0.456 (0.412, 0.500)	0.489	0.010	0.499
weight noise $s = 0.5$	5	0.480 (0.455, 0.504)	0.022 (0.014, 0.030)	0.003 (0.002, 0.003)	0.518 (0.497, 0.538)	0.529	0.024	0.545
weight noise $s = 0.003$	5	0.589 (0.568, 0.610)	0.047 (0.038, 0.055)	0.016 (0.012, 0.020)	0.745 (0.714, 0.777)	0.585	0.033	0.665

The loss-threshold attack in Table S15 is far weaker than LiRA even without privacy. Under privacy its within-stratum AUC ranges from 0.458 to 0.507, so its values below 0.5 do not arise from the mix of sources and classes in each split.

Table S15: Loss-threshold membership attack, AUC of the negative per-example loss with training images as members and test images as non-members. Within-stratum AUC averages the AUC computed inside each source and class stratum.

Mode	Partition	Regime	Pooled (full)	Stratum (full)	Pooled (clean)	Stratum (clean)	Figshare (clean)
central	natural	no DP	0.540	0.531	0.550	0.533	0.540
central	natural	clip only	0.494	0.479	0.510	0.478	0.488
central	natural	$\varepsilon = 8$	0.485	0.466	0.497	0.465	0.477
central	natural	$\varepsilon = 3$	0.493	0.474	0.499	0.474	0.492
central	natural	$\varepsilon = 1$	0.513	0.500	0.498	0.499	0.517
central	natural	patient $\varepsilon = 8$	0.511	0.507	0.546	0.507	0.505
central	natural	patient $\varepsilon = 3$	0.506	0.502	0.543	0.502	0.499
central	natural	patient clip only	0.505	0.477	0.539	0.477	0.482
central	natural	patient $\varepsilon = 16$	0.495	0.478	0.529	0.477	0.481
central	natural	patient $\varepsilon = 32$	0.501	0.481	0.536	0.481	0.488
central	natural	patient $\varepsilon = 64$	0.495	0.471	0.529	0.470	0.472
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.505	0.503	0.496	0.503	0.514
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.493	0.480	0.485	0.480	0.491
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.496	0.482	0.494	0.482	0.495
federated	Dir. 0.1, K=10	no DP	0.518	0.506	0.527	0.506	0.515
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.508	0.496	0.504	0.496	0.515
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.501	0.491	0.495	0.491	0.502
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.496	0.483	0.501	0.482	0.495
federated	Dir. 0.5, K=10	no DP	0.526	0.518	0.548	0.519	0.521
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.507	0.496	0.499	0.496	0.515
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.501	0.491	0.495	0.491	0.502
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.495	0.479	0.501	0.479	0.492
federated	Dir. 5.0, K=10	no DP	0.526	0.521	0.537	0.522	0.526
federated	IID, K=10	$\varepsilon = 1$	0.507	0.493	0.502	0.493	0.515
federated	IID, K=10	$\varepsilon = 3$	0.498	0.486	0.495	0.487	0.500
federated	IID, K=10	$\varepsilon = 8$	0.495	0.479	0.501	0.479	0.493
federated	IID, K=10	no DP	0.532	0.528	0.546	0.529	0.533
federated	natural	$\varepsilon = 1$	0.497	0.475	0.487	0.474	0.498
federated	natural	$\varepsilon = 3$	0.490	0.473	0.491	0.472	0.487
federated	natural	$\varepsilon = 8$	0.476	0.459	0.478	0.458	0.470
federated	natural	no DP	0.532	0.526	0.521	0.527	0.535
federated	natural	patient $\varepsilon = 8$	0.499	0.489	0.477	0.488	0.502
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.497	0.498	0.519	0.496	0.488
federated	natural	weight noise $s = 0.5$	0.506	0.497	0.487	0.497	0.514
federated	natural	weight noise $s = 0.003$	0.505	0.498	0.488	0.499	0.504

S6 Explanations

Table S16 reports localisation and faithfulness for Grad-CAM, Grad-CAM++ and Integrated Gradients across the privacy settings of central training. Grad-CAM shows the clearest decline as the privacy budget shrinks, Grad-CAM++ declines more gently, and Integrated Gradients reaches the highest pointing rate without privacy, a result that the randomisation check qualifies.

Table S16: Localisation and faithfulness of three attribution methods for central models on correctly classified Figshare test tumour images with expert masks, seed means. Lower deletion and higher insertion AUC indicate more faithful maps.

Regime	Method	Pointing	IoU top 15%	Energy in mask	Deletion	Insertion
no DP	Grad-CAM	0.197	0.087	0.032	0.290	0.604
no DP	Grad-CAM++	0.139	0.080	0.032	0.310	0.585
no DP	Int. Gradients	0.276	0.058	0.065	0.151	0.462
clip only	Grad-CAM	0.190	0.086	0.031	0.382	0.689
clip only	Grad-CAM++	0.152	0.084	0.032	0.389	0.676
clip only	Int. Gradients	0.200	0.054	0.057	0.214	0.479
$\varepsilon = 8$	Grad-CAM	0.156	0.076	0.029	0.406	0.715
$\varepsilon = 8$	Grad-CAM++	0.129	0.079	0.030	0.429	0.705
$\varepsilon = 8$	Int. Gradients	0.198	0.051	0.053	0.311	0.510
$\varepsilon = 3$	Grad-CAM	0.103	0.055	0.027	0.490	0.688
$\varepsilon = 3$	Grad-CAM++	0.098	0.069	0.028	0.494	0.708
$\varepsilon = 3$	Int. Gradients	0.172	0.046	0.048	0.353	0.531
$\varepsilon = 1$	Grad-CAM	0.089	0.032	0.027	0.480	0.526
$\varepsilon = 1$	Grad-CAM++	0.123	0.065	0.031	0.469	0.546
$\varepsilon = 1$	Int. Gradients	0.123	0.047	0.049	0.469	0.595
patient clip only	Grad-CAM	0.187	0.085	0.032	0.307	0.620
patient clip only	Grad-CAM++	0.115	0.077	0.032	0.330	0.590
patient clip only	Int. Gradients	0.239	0.055	0.060	0.155	0.396
patient $\varepsilon = 64$	Grad-CAM	0.127	0.058	0.029	0.290	0.555
patient $\varepsilon = 64$	Grad-CAM++	0.147	0.079	0.031	0.285	0.568
patient $\varepsilon = 64$	Int. Gradients	0.193	0.052	0.054	0.103	0.363
patient $\varepsilon = 32$	Grad-CAM	0.097	0.042	0.027	0.267	0.459
patient $\varepsilon = 32$	Grad-CAM++	0.129	0.075	0.029	0.251	0.521
patient $\varepsilon = 32$	Int. Gradients	0.168	0.047	0.048	0.102	0.322
patient $\varepsilon = 16$	Grad-CAM	0.069	0.032	0.025	0.269	0.375
patient $\varepsilon = 16$	Grad-CAM++	0.111	0.063	0.028	0.252	0.433
patient $\varepsilon = 16$	Int. Gradients	0.147	0.046	0.047	0.105	0.247
patient $\varepsilon = 8$	Grad-CAM	0.082	0.029	0.025	0.276	0.314
patient $\varepsilon = 8$	Grad-CAM++	0.102	0.047	0.028	0.238	0.363
patient $\varepsilon = 8$	Int. Gradients	0.206	0.049	0.051	0.163	0.248
patient $\varepsilon = 3$	Grad-CAM	0.056	0.023	0.018	0.345	0.300
patient $\varepsilon = 3$	Grad-CAM++	0.000	0.030	0.018	0.281	0.334
patient $\varepsilon = 3$	Int. Gradients	0.000	0.039	0.039	0.272	0.162

The randomisation check in Table S17 decides which method deserves interpretation. Integrated Gradients with a black baseline keeps a rank correlation of 0.683 to 0.712 with maps from a network whose weights were all re-initialised, so its maps mostly reflect the input image, whereas Grad-CAM correlations stay between -0.047 and 0.388. The check used one seed per setting and 64 images, which separates the two methods but does not rank privacy settings.

Table S17: Randomisation check. Spearman correlation between maps of the trained model and of a re-initialised network (central models, seed 1, 64 images), with all layers or only the last block and classifier re-initialised. Values near zero pass.

Regime	Grad-CAM, all layers	Grad-CAM, last block	IG, all layers	IG, last block
no DP	0.310	0.444	0.684	0.709
clip only	0.328	0.408	0.700	0.709
$\varepsilon = 8$	0.388	0.377	0.706	0.714
$\varepsilon = 3$	0.324	0.029	0.699	0.722
$\varepsilon = 1$	-0.047	0.309	0.703	0.728
patient clip only	0.166	0.477	0.683	0.705
patient $\varepsilon = 64$	0.232	0.491	0.694	0.719
patient $\varepsilon = 32$	0.309	0.403	0.703	0.726
patient $\varepsilon = 16$	0.035	-0.010	0.690	0.710
patient $\varepsilon = 8$	0.050	0.316	0.698	0.726
patient $\varepsilon = 3$	0.101	0.098	0.712	0.731

Table S18 tests for skull-border and background artefacts, adds Grad-CAM from the third residual block at twice the resolution and lists the cross-seed consistency used in the article. Grad-CAM places less mass on the brain rim and outside the brain than their area shares, so neither artefact explains the localisation results.

Table S18: Border-rim analysis, layer-3 Grad-CAM (8×8) and cross-seed consistency. Rim and outside-brain values are the share of Grad-CAM mass, next to the share of image area (chance).

Mode	Partition	Regime	Rim	Rim chance	Outside	Outside chance	L3 pointing	L3 IoU	Consistency
central	natural	no DP	0.175	0.334	0.261	0.405	0.192	0.055	0.700
central	natural	clip only	0.174	0.334	0.266	0.401	0.142	0.043	0.663
central	natural	$\varepsilon = 8$	0.176	0.334	0.280	0.398	0.123	0.044	0.485
central	natural	$\varepsilon = 3$	0.180	0.333	0.294	0.389	0.103	0.038	0.309
central	natural	$\varepsilon = 1$	0.180	0.351	0.362	0.443	0.051	0.027	0.235
central	natural	patient $\varepsilon = 8$	0.180	0.332	0.333	0.417	0.060	0.028	0.218
central	natural	patient $\varepsilon = 3$	0.176	0.308	0.289	0.329	0.056	0.009	0.234
central	natural	patient clip only	0.175	0.335	0.280	0.405	0.145	0.048	0.609
central	natural	patient $\varepsilon = 16$	0.181	0.335	0.318	0.406	0.066	0.030	0.225
central	natural	patient $\varepsilon = 32$	0.176	0.332	0.296	0.395	0.113	0.040	0.291
central	natural	patient $\varepsilon = 64$	0.174	0.332	0.299	0.399	0.156	0.049	0.368
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.163	0.358	0.406	0.452	0.056	0.024	0.291
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.176	0.335	0.290	0.395	0.045	0.021	0.364
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.174	0.334	0.274	0.395	0.120	0.037	0.489
federated	Dir. 0.1, K=10	no DP	0.173	0.334	0.269	0.399	0.101	0.037	0.687
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.193	0.359	0.407	0.462	0.051	0.026	0.302
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.176	0.331	0.258	0.383	0.082	0.035	0.451
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.173	0.333	0.258	0.390	0.131	0.043	0.513
federated	Dir. 0.5, K=10	no DP	0.174	0.335	0.261	0.401	0.161	0.049	0.725
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.188	0.359	0.406	0.458	0.049	0.022	0.312
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.176	0.330	0.258	0.382	0.088	0.034	0.456
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.173	0.333	0.255	0.389	0.141	0.043	0.523
federated	Dir. 5.0, K=10	no DP	0.173	0.335	0.264	0.404	0.157	0.048	0.703
federated	IID, K=10	$\varepsilon = 1$	0.192	0.360	0.416	0.462	0.048	0.022	0.303
federated	IID, K=10	$\varepsilon = 3$	0.175	0.330	0.251	0.381	0.089	0.038	0.476
federated	IID, K=10	$\varepsilon = 8$	0.173	0.332	0.254	0.388	0.140	0.043	0.526
federated	IID, K=10	no DP	0.173	0.334	0.262	0.403	0.155	0.048	0.705
federated	natural	$\varepsilon = 1$	0.174	0.335	0.287	0.401	0.074	0.032	0.356
federated	natural	$\varepsilon = 3$	0.175	0.331	0.251	0.386	0.124	0.044	0.532
federated	natural	$\varepsilon = 8$	0.175	0.333	0.280	0.399	0.118	0.041	0.467
federated	natural	no DP	0.174	0.335	0.267	0.405	0.118	0.041	0.657

Table S19 relates explanation quality to accuracy and noise. Both metrics correlate with clean balanced accuracy, and the noise multiplier adds no explanatory power once accuracy is in the fit.

Table S19: Explanation quality against clean balanced accuracy and the noise multiplier σ. Pearson r with accuracy, and least-squares coefficients of metric on accuracy and σ with the p-value of σ.

Metric	Models	r	p	b accuracy	b σ	p (σ)
Grad-CAM top-15Cross-seed consistency IoU	115	0.798	< 0.001	0.734	0.0049	0.153